

The economics of architecture

Gabriel M. Ahlfeldt

Humboldt University

Elisabetta Pietrostefani

University of Liverpool

Ailin Zhang

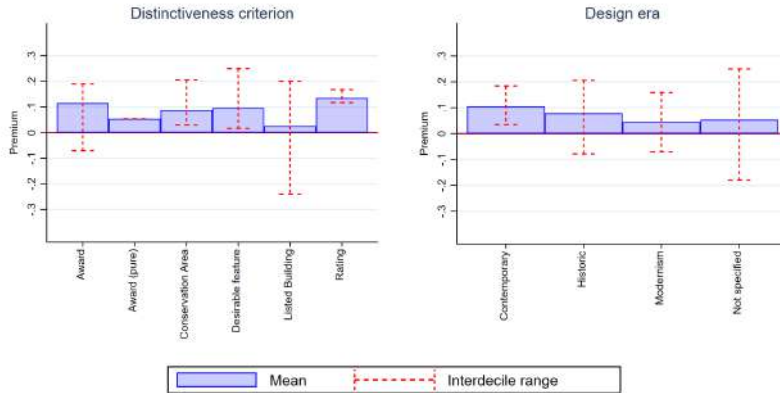
LSE

05/2026

Equilibrium

Internal design premium

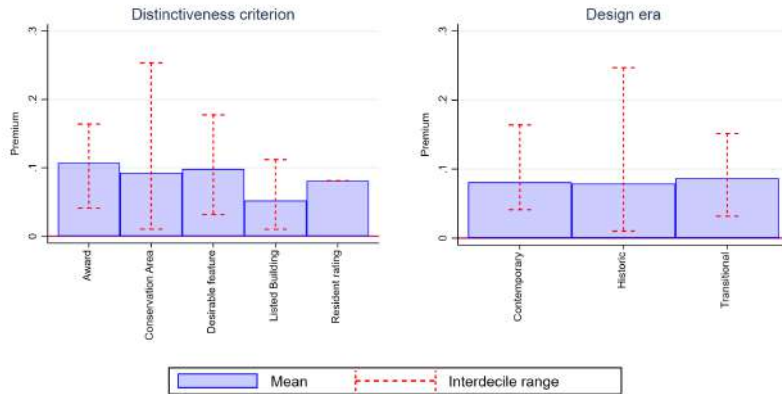
- ▶ Literature search reveals 68 estimates from 41 studies



	(1)	(2)	(3)
	Premium	Premium	Premium
Commercial	0.095*** (0.02)	0.101*** (0.02)	0.146*** (0.02)
Residential	0.077*** (0.02)	0.099** (0.04)	0.164*** (0.04)
Conservation Area		-0.02 (0.03)	
Listed Building		-0.08* (0.05)	
Award (pure)		-0.05 (0.04)	
Historic Style		0.010 (0.05)	
Weighted	No	No	Yes
Sample	All	All	Cohort control
r2	0.4	0.4	0.7
N	68	68	17

- ▶ Regression-based average
 - ▶ We weight observation by publication-year adjusted citation count
 - ▶ Most credible estimates are from studies that identify the distinctiveness effect within cohorts
 - ▶ **Internal design premium $\approx 15\%$**

- Literature search reveals 33 estimates from 28 studies: **Average $\approx 9\%$**



Design preferences

- ▶ **Internet survey** asking respondents to rate 10 buildings on a 0-100 scale
 - ▶ Also obtained ratings by "Starchitect" Stefano Boeri (Bosco Verticale, Milan)
- ▶ Structural estimation of **Fréchet shift and share parameters** using GMM: $\epsilon = 4$
 - ▶ Buildings rated highly by Starchitect also appreciated by respondents ($\rho = 0.8$)

Group	Mean rating	Variance	Shift A_b	Shape ε
All respondents, all buildings	52.72	480.88	42.73	4.04
People similar to Starchitect	51.34	416.81	39.53	4.06
People different from Starchitect	52.99	586.57	42.16	3.79
Buildings ranked high by Starchitect	65.63	486.86	55.89	4.69
Buildings ranked low by Starchitect	39.80	474.90	32.39	3.40

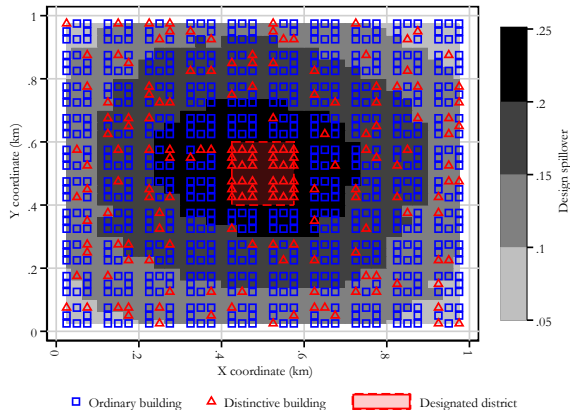
Notes: This table reports GMM estimates based on matching the first and second moments of the building-level rating distributions to those implied by a Fréchet model. The shape parameter ε governs preference heterogeneity, while A_b reflects the building-specific location shift. Groups are defined based on respondent-level or Starchitect-level ranking similarity.

13 / 43

Figure 1 consists of six line graphs arranged in a 2x3 grid, showing the impact of a 20% subsidy rate on various metrics. The x-axis for all graphs is 'Subsidy rate (%)' from 0 to 20. The y-axes represent different metrics. Each graph has a horizontal dashed line at 10% on the x-axis and a vertical dashed line at 10% on the y-axis.

- Top Left: # distinctive buildings** (y-axis: 160 to 200). The curve starts at approximately 164 at 0% subsidy, rises to about 178 at 10% subsidy, and continues to rise to approximately 194 at 20% subsidy.
- Top Middle: Subsidy per distinctive building (1,000,000)** (y-axis: 0 to 0.6). The curve is a straight line starting at (0,0) and ending at (20, 0.6).
- Top Right: Δ Expected (city-wide) utility (%)** (y-axis: 0 to 0.6). The curve is a straight line starting at (0,0) and ending at (20, 0.6).
- Bottom Left: Δ Aggregate gross developer profit (%)** (y-axis: 0 to 4). The curve starts at approximately 0.2 at 0% subsidy, rises to about 1.5 at 10% subsidy, and continues to rise to approximately 3.5 at 20% subsidy.
- Bottom Middle: Δ Aggregate net developer profit (%)** (y-axis: -30 to 0). The curve is a straight line starting at (0,0) and ending at approximately (20, -26).
- Bottom Right: Monetized benefit (% of income)** (y-axis: -0.1 to 0.1). The curve starts at 0 at 0% subsidy, rises to a peak of approximately 0.1 at 10% subsidy, and then falls back to 0 at 20% subsidy.

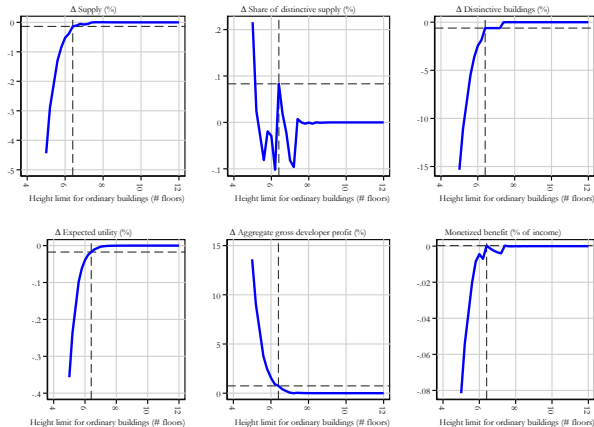
- Subsidy gradients Sensitivity



- ▶ Can **force developers** to solve the coordination problem
 - ▶ All buildings in a district must be distinctive
 - ▶ Related policy: Conservation areas
- ▶ Can generate **benefits when the distinctive district is small**
 - ▶ Distinctive buildings +4.3%
 - ▶ Expected utility +0.003% (city-wide)
 - ▶ Developer profits +0.018%
- ▶ If it is too large, the effect is negative due to diminishing returns

Gradient: Large district

FAR incentives



- ▶ **FAR incentives only work under constraints for ordinary buildings**
 - ▶ Introduce height limit for ordinary buildings
 - ▶ No limit for distinctive buildings
- ▶ Generally, binding **height limits depreciate welfare**
 - ▶ Non-binding height limits do not affect distinctive supply
- ▶ **Knife-edge welfare gain**
 - ▶ At 6.4-floor height limit
 - ▶ hard to hit in practice
 - ▶ Tighter constraints depreciate aggregate welfare and transfer welfare from workers to landowners

Ordinary



London



Barcelona



Bath



Milan



Distinctive

Context

Developers: Technology and profits

- ▶ One developer per parcel i
 - ▶ Chooses building design $d \in \{\text{distinctive}, \text{ordinary}\}$
 - ▶ Chooses height h_i^d on parcel land $\bar{K}_i$
- ▶ Revenue:
 - ▶ $Q_i^d h_i^d \bar{K}_i$
 - ▶ Q_i^d is the unit price of floor space
- ▶ Construction cost:
 - ▶ $\bar{C}(h_i^d)^{1+\theta} \exp(\delta_i^d - t_i^d) \bar{K}_i$
 - ▶ θ governs how construction costs rise with height
 - ▶ δ_i^d is a design-specific cost shifter
 - ▶ t_i^d is a construction-cost subsidy
- ▶ Profit:
 - ▶ $\pi_i^d = \left[Q_i^d h_i^d - \bar{C}(h_i^d)^{1+\theta} e^{\delta_i^d - t_i^d} \right] \bar{K}_i$

Developers: Height and design choice

- ▶ Profit-maximising height:

$$\text{▶ } h_i^{d*} = \left(\frac{Q_i^d}{(1+\theta)\bar{c}e^{\delta_i^d - t_i^d}} \right)^{1/\theta}$$

- ▶ Height regulation:

$$\text{▶ } \tilde{h}_i^d = \min(h_i^{d*}, \bar{h}_i^d)$$

- ▶ $\bar{h}_i^d$ is the design-specific height limit

- ▶ Design choice:

- ▶ If $G_i = \text{distinctive}$: planner requires distinctive design

- ▶ Else: developer chooses

$$\text{▶ } \tilde{d}_i = \arg \max_{d \in \mathcal{D}} \pi_i^d$$

- ▶ Floor-space supply:

$$\text{▶ } H_i^d = \tilde{h}_i^d \mathbf{1}(\tilde{d}_i = d) \bar{K}_i$$

- ▶ Key mechanism:

- ▶ Developers internalize own rents, but not spillovers on neighbouring parcels

Welfare

- ▶ Two welfare components:
 - ▶ Worker utility: $\mathbb{E}[U]\bar{N}$
 - ▶ Developer profits:
 - ▶ $\Pi = \sum_{i,d} \pi_i^d \mathbb{1}(\tilde{d}_i = d) + \bar{\pi}^{outside}$
- ▶ Expected utility:
 - ▶ $\mathbb{E}[U] = \Gamma\left(\frac{\varepsilon-1}{\varepsilon}\right) \left(\sum_{u \in \mathbf{C}} (\tilde{V}^u)^\varepsilon\right)^{1/\varepsilon}$
- ▶ Welfare metric:
 - ▶ Utility changes converted into income-equivalent units
 - ▶ Cobb-Douglas structure implies utility scales proportionally with income
- ▶ Policy evaluation:
 - ▶ $\Delta W = \frac{\mathbb{E}[U]^1 - \mathbb{E}[U]^0}{\mathbb{E}[U]^0} \bar{w} \bar{N} + \Delta \Pi$
- ▶ Subsidy cost:
 - ▶ $S = \sum_{i,d} \mathbb{1}(\tilde{d}_i = d) \bar{C}(h_i^d)^{1+\theta} \bar{K}_i \left[e^{\delta_i^d} - e^{\delta_i^d - t_i^d} \right]$

Equilibrium: Spatial equilibrium conditions

- ▶ Workers are mobile:
 - ▶ Across neighbourhoods via outside option (constant utility)
 - ▶ Across submarkets within neighbourhood (constant expected utility)
- ▶ Developers are competitive:
 - ▶ Take rents and spillovers as given
 - ▶ Choose profit-maximising design and height
- ▶ Floor-space markets clear at parcel level:
 - ▶ $f_i^d N_i^d = H_i^d$
- ▶ Submarket populations are consistent with discrete choice:
 - ▶ $\mu^d = \frac{1}{N} \sum_i N_i^d$
 - ▶ $\mu^d = \frac{(\tilde{V}^d)^\varepsilon}{\sum_{u \in \mathcal{C}} (\tilde{V}^u)^\varepsilon}$
- ▶ Spatial equilibrium:
 - ▶ Rents adjust such that demand equals supply across submarkets

Equilibrium: Recursive structure

- ▶ Conditional on:
 - ▶ $\{\tilde{d}_i, Q_i^d, \tilde{V}^{distinctive}, \tilde{V}^{ordinary}\}$
- ▶ Workers:
 - ▶ Housing demand: $f_i^d = (1 - \alpha) \frac{\bar{w}}{Q_i^d}$
 - ▶ Choice probabilities: $\mu^d = \frac{(\tilde{V}^d)^\varepsilon}{\sum_{u \in \mathbf{C}} (\tilde{V}^u)^\varepsilon}$
- ▶ Developers:
 - ▶ Optimal height: $h_i^{d*} = \left(\frac{Q_i^d}{(1+\theta)\bar{c}e^{\delta_i^d - t_i^d}} \right)^{1/\theta}$
 - ▶ Feasible height: $\tilde{h}_i^d = \min(h_i^{d*}, \bar{h}_i^d)$
- ▶ Implied equilibrium objects:
 - ▶ Spillovers D_i , amenities B_i^d , supply H_i^d
 - ▶ Population N_i^d , utility $\mathbb{E}[U]$, profits Π

Equilibrium: Fixed-point problem

- ▶ Unknown equilibrium objects:
 - ▶ Design choices $\tilde{d}_i$
 - ▶ Rents Q_i^d
 - ▶ Submarket utilities $\{\tilde{V}^{distinctive}, \tilde{V}^{ordinary}\}$
- ▶ Submarket utilities pinned down by:
 - ▶ Population consistency: $\mu^d = \frac{1}{N} \sum_i N_i^d = \frac{(\tilde{V}^d)^\varepsilon}{\sum_{u \in \mathbf{C}} (\tilde{V}^u)^\varepsilon}$
- ▶ Design choices pinned down by:
 - ▶ Profit maximisation: $\tilde{d}_i = \arg \max_{d \in \mathbf{D}} \pi_i^d$
- ▶ Rents pinned down by:
 - ▶ Parcel-level market clearing: $f_i^d N_i^d = H_i^d$
- ▶ Numerical solution:
 - ▶ Fixed point because spillovers affect both rents and design choices
 - ▶ Avoid multiplicity via convexification (heterogenous design cost)

Survey Examples I



40 (23)



48 (22)



29 (19)



39 (23)



61 (25)



75 (21)



72 (19)



76 (19)



50 (24)



42 (21)



37 (21)



55 (25)



73 (19)



58 (23)

[Back to Preferences](#)

[back](#)

[More](#)



34 (21)



46 (24)



35 (20)



62 (22)



66 (24)



57 (23)

[Back to Preferences](#)

[back](#)

Survey: Fréchet taste heterogeneity

- ▶ Survey data:
 - ▶ Respondents rate buildings on a 0–100 scale
 - ▶ Ratings interpreted as latent taste draws a_{ib}
- ▶ Structural assumption:
 - ▶ a_{ib} follows a Fréchet distribution
 - ▶ $F(a) = \exp(-(a/A_b)^{-\varepsilon})$
- ▶ Parameters:
 - ▶ A_b : building-specific attractiveness
 - ▶ ε : taste heterogeneity across respondents
- ▶ Theoretical moments:
 - ▶ $\mathbb{E}[a_b] = A_b \Gamma(1 - \frac{1}{\varepsilon})$
 - ▶ $\text{Var}(a_b) = A_b^2 \left[\Gamma(1 - \frac{2}{\varepsilon}) - \Gamma(1 - \frac{1}{\varepsilon})^2 \right]$

GMM Estimation

- ▶ Empirical moments:
 - ▶ $\hat{\mu}_b = \frac{1}{n_b} \sum_{i=1}^{n_b} a_{ib}$
 - ▶ $\hat{\sigma}_b^2 = \frac{1}{n_b-1} \sum_{i=1}^{n_b} (a_{ib} - \hat{\mu}_b)^2$
- ▶ Moment conditions:
 - ▶ $g_b^\mu(\theta) = \hat{\mu}_b - A_b \Gamma \left(1 - \frac{1}{\varepsilon}\right) = 0$
 - ▶ $g_b^\sigma(\theta) = \hat{\sigma}_b^2 - A_b^2 \left[\Gamma \left(1 - \frac{2}{\varepsilon}\right) - \Gamma \left(1 - \frac{1}{\varepsilon}\right)^2 \right] = 0$
- ▶ Identification:
 - ▶ 20 buildings $\Rightarrow$ 40 empirical moments
 - ▶ Mean and variance for each building
 - ▶ 21 parameters (ε and 1 A_b per building) $\Rightarrow$ overidentified system
- ▶ Estimation method: One-step GMM with identity weighting matrix

[Back to Preferences](#)

SMM: Calibration targets

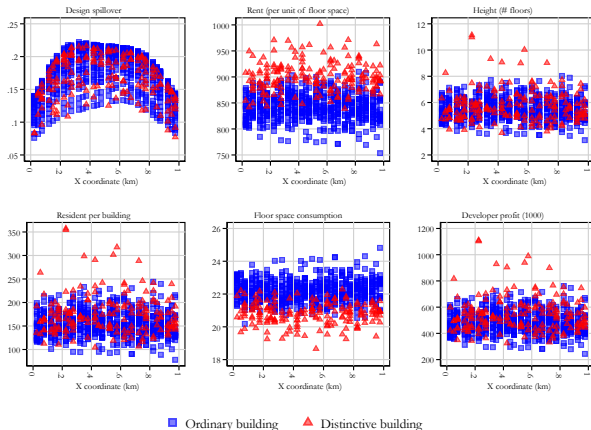
- ▶ Parameters to calibrate:
 - ▶ Preference for distinctive design: $A^{d=\text{distinctive}}$
 - ▶ Average distinctive design cost: $\bar{\delta}^{d=\text{distinctive}}$
- ▶ Distinctive design cost: Common component plus shock from normal distribution
 - ▶ $\delta_i^{d=\text{distinctive}} = \bar{\delta}^{d=\text{distinctive}} \left(1 + \sigma \frac{\tilde{\delta}_i}{\bar{\delta}^{d=\text{distinctive}}} \right)$
- ▶ Empirical target moments:
 - ▶ Internal design premium: $\hat{a} = 15\%$
 - ▶ Share of distinctive buildings: $\hat{s} = 20\%$
- ▶ SMM objective:
 - ▶ Match model-implied premium and distinctive share
 - ▶ Internal premium identifies preferences
 - ▶ Distinctive share identifies costs

SMM: Numerical procedure

- ▶ Inner loop:
 - ▶ Solve developer design choices and heights
 - ▶ Iterate until no parcel switches design
- ▶ Middle loop:
 - ▶ Update utilities and population shares
 - ▶ Clear housing markets across submarkets
- ▶ Outer loop:
 - ▶ Update structural parameters: $A^{d=\text{distinctive}}$,
 - ▶ $\bar{\delta}^{d=\text{distinctive}}$
- ▶ Numerical objective:
 - ▶ Minimize weighted distance between:
 - ▶ Simulated and empirical internal premium
 - ▶ Simulated and empirical distinctive share
- ▶ Main calibration:
 - ▶ $A^{d=\text{distinctive}} = 0.772$, $\bar{\delta}^{d=\text{distinctive}} = 0.219$

[Back to Quantification](#)

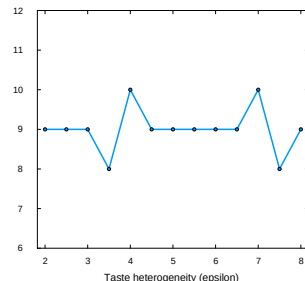
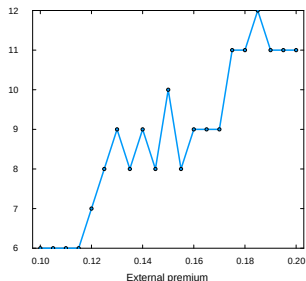
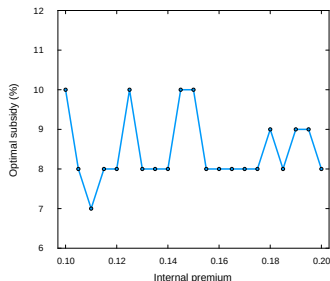
Subsidy Gradients



- Effects across subsidy levels:
 - Distinctive building share rises smoothly
 - Rents increase, but at diminishing rate
- Margins of adjustment:
 - Utility gain from spillovers
 - Developer profits vs. tax burden
- Welfare profile:
 - Concave, consistent with downward-sloping demand
 - Peak defines Pigovian optimum

[Back to Pigovian subsidy](#)

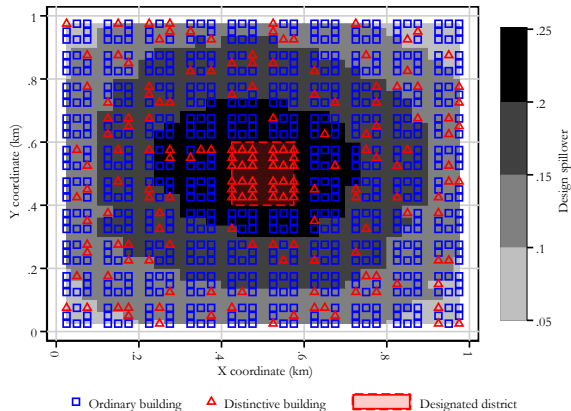
Sensitivity of the optimal subsidy



- ▶ The optimal subsidy remains remarkably stable across parameter values (6–12%)
- ▶ Sensitivity is greatest with respect to the **external design premium**, which directly measures the size of the design spillover

[Back to Pigovian subsidy](#)

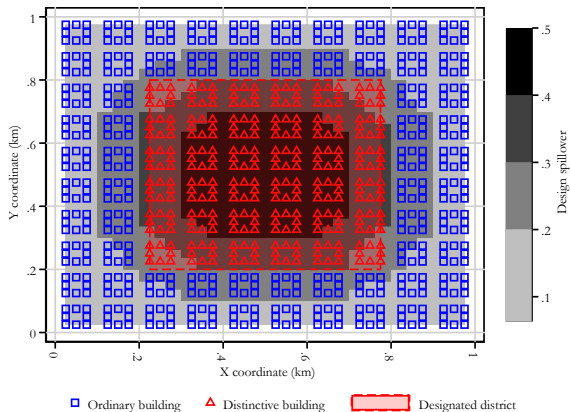
Mandatory District Map



- Visualizes the location of the mandatory design district
- Shows spatial clustering of distinctive buildings
- Useful for interpreting spillover and coordination effects

[Back to Mandatory District](#)

Mandatory District Map

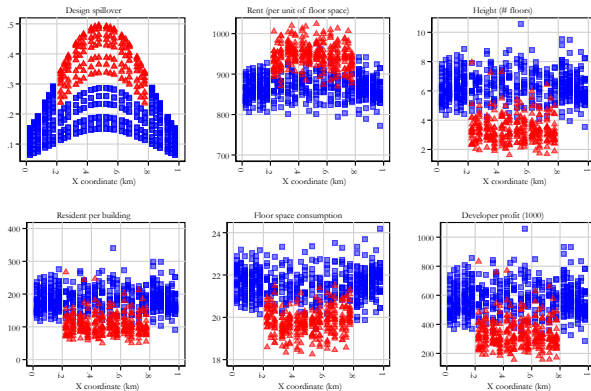


- ▶ All distinctive buildings now in the district
- ▶ Many more distinctive buildings
- ▶ Profitable distinctive buildings **crowded out**

[Back to Mandatory District](#)

41 / 43

Mandatory District: Large-Scale Scenario

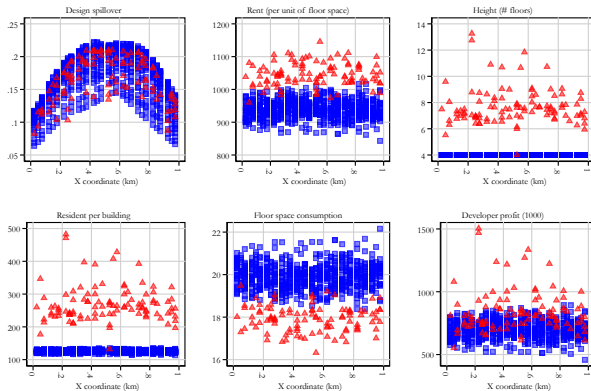


■ Ordinary building ▲ Distinctive building

- ▶ When the mandatory district is large:
 - ▶ Distinctive share saturates
 - ▶ Returns diminish, congestion effects rise
- ▶ Welfare impact turns negative
 - ▶ Utility and profits decline
- ▶ Policy trade-off:
 - ▶ Small districts solve coordination
 - ▶ Large ones impose inefficiency

[Back to Mandatory District](#)

FAR incentives: Gradients



■ Ordinary building ▲ Distinctive building

- ▶ With binding height constraints on ordinary buildings
 - ▶ Heights of distinctive buildings increase
 - ▶ Positive distinctive design premium persists
 - ▶ Some distinctive buildings generate large developer profits

Back to FAR incentives